ATTACHMENT 1

DOCUMENT DATE OCT 31, 2022

CONDITIONAL MANAGEMENT PLAN (CMP)

BETWEEN THE:

Canadian Food Inspection Agency (CFIA),
Fisheries and Oceans Canada (DFO),
Environment and Climate Change Canada (ECCC)
British Columbia Ministry of Environment and Climate Change Strategy, Environmental Protection Division (ENV), and
The Town of Ladysmith

(the “Participants”)

FOR THE MANAGEMENT OF SHELLFISH HARVESTING IN CONDITIONALLY CLASSIFIED HARVEST AREAS ADJACENT TO THE WASTEWATER TREATMENT PLANT LOCATED AT:

245A Oyster Cove, Town of Ladysmith, BC
(the “Ladysmith WWTP”)

INTRODUCTION

The Canadian Shellfish Sanitation Program (CSSP) is delivered by three federal institutions, Fisheries and Oceans Canada (DFO), Environment and Climate Change Canada (ECCC) and the Canadian Food Inspection Agency (CFIA) (hereinafter collectively referred to as the “CSSP partners”). The CSSP provides reasonable assurance that bivalve molluscan shellfish are safe for consumption, thus supporting the industry and providing confidence for Canadians and export markets. This Conditional Management Plan (CMP) between the CSSP partners, the Province of British Columbia Ministry of Environment and Climate Change Strategy, Environmental Protection Division (ENV) and, the Municipality/Wastewater Treatment Plant (WWTP) Operators outlines the roles and responsibilities of the Participants in the event of a trigger event (described in Appendix E of this CMP) at the Ladysmith WWTP.

The present CMP does not affect legal requirements existing under Federal or Provincial statutes. For greater certainty, where there occurs a deposit of a deleterious substance out of the normal course of events in water frequented by fish, or a serious and imminent danger thereof, the person that causes or contributes to the deposit (or danger thereof) has a legal obligation to report such occurrences to an inspector designated pursuant to subsection 38(1) of the *Fisheries Act (R.S.C., 1985, c. F-14),* to a fishery officer, or to any authority as is prescribed by regulation under subsection *38(4)* of the *Fisheries Act (R.S.C., 1985, c. F-14)*.

This CMP does not affect the legal rights and responsibilities of the Municipality, as a local government under the *Community Charter [SBC 2003] CHAPTER 26* and the *Local Government Act [RSBC 2015] CHAPTER 1*, or as a WWTP operator subject to all applicable regulatory licences and permits, but reflects the Municipality’s commitment to assist the CSSP partners in providing reasonable assurance that bivalve shellfish harvested adjacent to the Ladysmith WWTP area are safe for human consumption.

DURATION OF THE CONDITIONAL MANAGEMENT PLAN

This CMP shall be renewed on the date of signing by each of the Participants and expires on January 31, 2026 and is subject to each participant signing this CMP prior to coming into force.

1 PURPOSE AND SCOPE

Full cooperation of the Participants is required to achieve timely detection and notification of WWTP trigger events, along with the necessary response actions to ensure continued food safety of harvested bivalves. This initiative stems from the CSSP and the requirements related to Conditional Area Management (refer to CSSP Manual, a copy of which is located at http://www.inspection.gc.ca/food/sfcr/food-specific-requirements-and-guidance/fish/canadian-shellfish-sanitation-program/eng/1527251566006/1527251566942). Conditional Area Management will be supplemented by Preventive Control Plans implemented by Safe Food for Canadians licensed operators.

This CMP provides enhanced management of the bivalve molluscan shellfish harvest waters adjacent to the Ladysmith WWTP as described in Section 4.

The area described in Section 4 of this CMP has been classified based upon the area hydrographical data and the performance characteristics of the Ladysmith WWTP. Any such discharges outside the normal operations of the Ladysmith WWTP, including rare events such as catastrophic failures at the WWTP or ruptures in the collection system that result in wastewater discharges to the marine environment that are not specifically listed under the trigger event definition in this CMP, remain subject to reporting under subsection 38(4) of the *Fisheries Act (R.S.C., 1985, c. F-14)*, with any subsequent closures of shellfish harvesting areas addressed under Section B, 6.2 of the CSSP Manual. A classification map of the harvest area is provided in Appendix A.

2 BACKGROUND

2.1 Conditional Area Management

The CSSP Manual outlines the authorities (statutes and regulations), policies, procedures, and activities governing the control of shellfish growing areas, and the harvesting, processing and distribution of shellfish.

Shellfish harvest areas outlined in this CMP that are subject to intermittent microbiological contamination may be classified as conditionally restricted areas. If the conditions set out in this CMP cannot be met, the CSSP partners will determine whether the area will be reclassified as prohibited.

2.2 WWTP description

The Ladysmith WWTP operated under provincial authorization (PE-00120) issued by the ENV until April 4, 2022. A Liquid Waste Management Plan (LWMP) for the Town of Ladysmith was approved on April 2, 2013, and Operational Certificate 110161 was issued by the provincial authority on April 4, 2022 to replace permit PE 00120. The authorized discharges are described in Section 5.1.2.

The authorized works are influent screening, primary treatment using the Salsness Filter process; secondary (biological) treatment using the moving bed bioreactor (MBBR) process; dissolved air floatation (DAF) for separation of biological solids; disinfection using sodium hypochlorite followed by the dechlorination using calcium thiosulphate); 710 mm diameter outfall extending 850 m (48°59.00′ north latitude and 123°47.46' west longitude) from the mean low water to a depth of 17.9 m below low water; and sludge handling and dewatering facilities located on site.

The system is designed with four potential overflow points described as follows:

- Salsness filter failure: Effluent would be diverted to the old spirogester with a 700 m^3 capacity, and then be directed to the chlorine contact chamber then dechlorinated. If the failure results in flows that are less than that of the capacity of the spirogester, this effluent will be pumped back into the head of the treatment facility when capacity exists. This would result in no trigger event. If the chlorine contact chamber is full it will be directed (manually) into the outflow line. The contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.
- MBBR failure (overflow): Effluent would be diverted to the chlorine contact chamber then dechlorinated. This effluent would have passed through the Salsness filters so has received primary treatment. If the chlorine contact chamber is full it will be directed (manually) into the outflow line. The chlorine contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.
- Spirogester overflow: If the spirogester is full, the system will back up into the overflow weir and into the outfall. A bypass flow of 50 m^3/day will impact shellfish resources in Ladysmith Harbour.
- Wet Well pump failures and/or over capacity (overflow): Effluent would be diverted to the spirogester with a 700 m^3 capacity, and then be directed to the Chlorine contact chamber then dechlorinated. If the failure results in flows that are less than that of the capacity of the spirogester, this effluent will be pumped back into the head of the treatment facility when capacity exists. This would result in no trigger event. If the chlorine contact chamber is full it will be directed (manually) into the outflow line. The chlorine contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.

Collection system lift stations are not designed with a bypass discharge.

2.3 Description of the Shellfish Fisheries (including aquaculture) within the CMP Area

Significant bivalve shellfish harvest takes place near the Ladysmith WWTP within conditionally restricted classification areas. For all types of potential use, fisheries and aquaculture licence conditions prohibit harvest where notice is given of biotoxin or other contamination such as trigger events from the Ladysmith WWTP. Where not closed by Prohibition Order, Variation Orders open commercial bivalve fisheries in non-contaminated areas for specified areas and times.

Within the conditionally restricted area, where appropriately licensed, wild and aquaculture harvest may occur. In this area, there are six shellfish aquaculture tenures issued by the Province and licensed by DFO.

Pacific Oyster and clam relay and commercial intertidal clam depuration, may occur in marginally contaminated areas that are closed by Prohibition Order; these activities must be reviewed by ECCC and CFIA and authorized by licences issued by DFO under the *Management of Contaminated Fisheries Regulations (SOR/90-351).*

3 ROLES AND RESPONSIBILITIES

The roles and responsibilities of each of the Participants are specified in Sections 3 and Sections 5 to 8 of this CMP. Additional responsibilities for the CSSP partners can be found in Section E, 13 of the CSSP Manual.

The roles and responsibilities of the ENV and the Town of Ladysmith are as follows:

The ENV is responsible for regulating municipal WWTP's. The ENV will:

a) verify through annual report reviews, and any other communications with the Ladysmith WWTP Chief Operator or designate that trigger events, are reported by the methods described in this CMP, as they occur.
b) provide an annual summary report of the results of compliance with provincial authorization requirements and confirmed trigger event reporting as defined in Sections 5.1 and 5.2 of this CMP by February 15th for the previous calendar year, including recommendations for changes to this CMP.

The WWTP Chief Operator or designate of the Town of Ladysmith is responsible for the operation of the Ladysmith WWTP located at 245A Oyster Cove, Ladysmith, BC. The WWTP Chief Operator or designate of the Town of Ladysmith will:

a) where feasible, maintain a continuous monitoring system by which trigger events described in Section 5.1 can be detected in a timely manner, and improve upon that monitoring system;
b) immediately notify the DFO Radio Room and Emergency Management BC (EMBC) verbally by telephone of any planned or unplanned changes in operations of the municipality's wastewater collection and treatment facility which may or has resulted in a trigger event as per Section 5.1;

c) advise CFIA, ECCC and DFO in writing by email using the online Wastewater Treatment Plant Reporting Tool when the trigger event has occurred or has ceased**.** Alternatively, advise the DFO Radio Room and EMBC by telephone as per Section 5.2. This notification is the initial step in the re-opening criteria process;

d) in the event that a trigger event is detected as described in 5.1, WWTP Chief Operator or designate will implement the Notification procedures as described in section 5.2 of this CMP. This includes an initial call to EMBC and the DFO Radio Room, and then using the web-based "Wastewater Treatment Plant Reporting Tool" to submit a record of the trigger event for CMP purposes;

e) maintain up-to-date records of the operations and maintenance of the wastewater collection and treatment facilities as per the requirements of the operating certificate 110161 issued by ENV;

f) provide a copy of the results of routine final effluent analysis to ENV and ECCC, described in the operating permit issued by ENV;

g) update the record of the discharge event using the "Wastewater Treatment Plant Reporting Tool" when the trigger event has terminated, which will implement Notification procedures that will advise the Participants that the trigger event has ceased ("END of Discharge Event" email);

h) provide an annual report of the results of activities listed above to the ENV and ECCC for the 12 month period starting January 1st and finishing December 31st of each year. The report must contain at minimum the occurrence trigger events including dates, estimates of discharges and records of the notifications given to DFO and EMBC, and notifications given to DFO when the event ceased. The report must be submitted by January 15th of the following year and may contain recommendations for changes to this CMP if any are necessary.

i) as of April 1, 2022 in the event that a trigger event occurs outside of regular business hours (Monday-Friday between the hours of 08:00-16:30 and statutory holidays) the DFO standby officer on duty is to be immediately notified, in replacement of the DFO Radio Room. A standby calendar containing the name and contact information for the officer on standby has been provided in the event of any after-hours emergencies.

4 DESCRIPTION OF THE SANITARY CLOSURES, CONDITIONALLY RESTRICTED CLASSIFIED AREA(S)

This CMP deals specifically with the harvesting of shellfish in the conditionally restricted area(s) described below.

4.1 Sanitary Closure 17.1, including:
The waters and intertidal foreshore of Ladysmith Harbour lying inside a line drawn from Sharpe Point at 48°58.88' north latitude and 123°46.10' west longitude, thence southeasterly to a point on land southeast of Boulder Point at the foot of South Oyster

School Road at 48°57.49' north latitude and 123°45.35' west longitude. [Sanitary Closure Map 17.1] [NAD 83]

4.2 Conditionally Restricted Areas within Sanitary Closure 17.1, including:

That portion of Ladysmith Harbour inside a line that starts at an unnamed point of land at 48°58.97' north latitude and 123°46.37' west longitude northwest of Sharpe Point, thence westerly to a point in water at 48°59.02' north latitude and 123°47.03' west longitude, thence northerly to a point on land in Sibell Bay at 48°59.57' north latitude and 123°46.88' west longitude, thence following the shoreline along the high water mark to the point of commencement. [NAD 83]

And:

That portion of Ladysmith Harbour inside a line that starts at a point of land at Hunter Point at 48°59.51' north latitude and 123°47.18' west longitude, thence southwesterly to a point on land on the islet off the westernmost end of the easternmost Dunsmuir Island at 48°59.45' north latitude and 123°47.28' west longitude, thence following the northern shoreline high water mark of the westernmost Dunsmuir Island to the westernmost point at 48°59.59' north latitude and 123°47.73' west longitude, thence easterly to a point in water at 48°59.61' north latitude and 123°47.51' west longitude, thence northerly to an unnamed point of land at 48°59.80' north latitude and 123°47.78' west longitude, thence following the shoreline along the high water mark to the point of commencement. [NAD 83]

And:

That portion of Ladysmith Harbour inside a line that starts at a point of land at 48°58.71' north latitude and 123°47.92' west longitude, thence northerly to a point in water at 48°58.94' north latitude and 123°47.77' west longitude, thence northwesterly to a point in water at 48°59.12' north latitude and 123°48.04' west longitude thence westerly to a point on land at 48°59.13' north latitude and 123°48.38' west longitude, thence following the shoreline along the high water mark to the point of commencement. [NAD 83]

4.3 Boundaries and Orders may Change

Classification boundaries and Prohibition Orders may be amended during the term of this CMP as required according to on-going sampling data and advice from ECCC and/or CFIA and are also subject to any overlapping prohibited areas such as 125 m radius around floating living accommodations and 300 m around outfalls, as per the MCFR. See Appendix A.

5 DETECTION / NOTIFICATION / RESPONSE TO A TRIGGER EVENT

As defined in the CSSP Manual, an effective regime for the detection, notification, and response to disruptions in the normal operation of a WWTP is a prerequisite to the harvest of shellfish in the conditionally restricted areas described in Section 4 of this CMP.

5.1 Detection

The Town of Ladysmith WWTP Chief Operator or designate will maintain a continuous monitoring system by which trigger events set forth in this CMP can be detected in a timely manner. The importance of continuous monitoring systems is recognized by the Town of Ladysmith and they will continue to evaluate and upgrade systems as part of their capital planning process.

The conditionally restricted areas described in Section 4 of this CMP will be closed to the harvesting of molluscan shellfish in response to any trigger event that results in a discharge to the marine environment that has the potential to pose a contamination risk to shellfish beyond the existing Sanitary Closure 17.1 boundary, as defined in further detail in Section 5.1.3.

The Town of Ladysmith WWTP Chief Operator or designate must be able to detect such discharges, and to make the appropriate notifications.

Trigger events include, but are not limited to, the following conditions:

- Daily effluent discharge volume exceeding the WWTP plant design capacity of 14,400 m^3/day
- WWTP disruptions that result in more than 50 m^3/day of untreated effluent being discharged via the bypass outfall
- WWTP disruptions that result in untreated or partially treated effluent discharge that fails to meet ENV authorized effluent disinfection characteristics
- Bypasses identified but not limited to those identified in Section 5.1.3 that result in effluent discharges not in receipt of secondary treatment
- Collection system failures that result in an untreated wastewater discharge to the marine environment

Rare events, such as catastrophic failures or collection system ruptures, that result in sewage discharges to the marine environment and are not specifically listed under the trigger event definition in this CMP, remain subject to reporting under subsection 38(4) of the *Fisheries Act (R.S.C., 1985, c. F-14),* as well as the CSSP Emergency Events Section B, 6.2 of the CSSP Manual.

5.1.1 Description of the normal operating requirements (performance standards or values permitted by provincial regulators)

The plant operated under ENV authorization (PE-00120) issued by the ENV until April 4, 2022. A LWMP for the Town of Ladysmith was approved on April 2, 2013, and Operational Certificate 110161 issued by the provincial authority, replaced permit PE-00120. The characteristics of the authorized discharges pertaining to the CMP are as follows:

The maximum rate at which effluent may be discharged through the Ladysmith WWTP is 15,500 m^3/day, 365 days per year. Flow in excess of the treatment plant capacity is screened and discharged to the outfall. The characteristics of the discharge must be equivalent to or better than: Carbonaceous 5-day Biochemical Oxygen Demand, Maximum: 45 mg/L; Total Suspended Solids, Maximum: 45 mg/L; Fecal Coliforms, Maximum: 14 CFU/100mL or MPN/100 mL at the Initial Dilution Zone boundary.

The effluent shall be dechlorinated prior to discharge to reduce the total chlorine residual below 0.02 mg/l.

5.1.2 Description of scenarios that are reasonably likely to occur resulting in a trigger event (lack of disinfection, bypass, power failure, overflow of lift stations that could impact the area, presence of a hazardous substance such as oil or gas, others)

Bypass: The system is designed with an overflow bypass as follows:

• Salsness filter failure: Effluent would be diverted to the old spirogester with a 700 m^3 capacity, and then be directed to the chlorine contact chamber for dechlorination. If the failure results in flows that are less than that of the capacity of the spirogester, this effluent will be pumped back into the head of the treatment facility when capacity exists. This would result in no trigger event. If the chlorine contact chamber is full it will be directed (manually) into the outflow line. The chlorine contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.

• MBBR failure (overflow): Effluent would be diverted to the chlorine contact chamber then dechlorinated. This effluent would have passed through the Salsness filters so has received primary treatment. The chlorine contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.

• Wet Well pump failures and/or over capacity (overflow): Effluent would be diverted to the old spirogester with a 700 m^3 capacity, and then be directed to the chlorine contact chamber then dechlorinated. If the failure results in flows that are less than that of the capacity of the spirogester, this effluent will be pumped back into the head of the treatment facility when capacity exists. This would results in no trigger event. If the chlorine contact chamber is full it will be directed (manually) into the outflow line. The chlorine contact chamber will have a SCADA alarm for high level, if this system fails the chlorine contact chamber could overflow into the harbour which will impact shellfish resources in Ladysmith Harbour.

• Spirogester overflow: If the spirogester is full the system will back up into the overflow weir and into the outfall. A bypass flow of 50 m^3/day will impact shellfish resources in Ladysmith Harbour.

Power Failure: WWTP function will continue in the case of a power failure due to a dedicated permanent generator.

Disinfection: Disinfection levels are maintained during power failures due to a dedicated permanent generator.

Lift stations: Lift stations have been designed with a 10 hour holding capacity and high level alarms. The town has 7 lift stations, in the event of a power failure the following will occur:

- Swettenham Pl. and Parks Pl. have onsite standby generators
- Ludlow and Dogwood Dr. would be connected to portable generators.
- Transfer Beach would be shut down.
- Sandy Beach and Gill Rd. have no standby generators or ability to be hooked to a portable generator, so would rely on pumper trucks.

5.1.3 Description of how each of the trigger event types noted above are detected (Supervisory Control and Data Acquisition (SCADA), visual, others).

Detection of trigger events during and after normal work hours, as they currently exist include the following:

a) Power failure and other process failures are connected to a SCADA system that notifies the operator on-call. Response time is 0.25 hours. This capability is available 24 hours a day 7 days a week.
b) Sodium hypochlorite is metered into the contact chamber. If a power failure occurs the system operates as normal due to the backup generator. The Town conducts daily tests to confirm residual levels.
c) Backup generators power two lift stations and the WWTP including the disinfection system.

Lift station wells are alarmed with an auto dialer and have enough capacity for 10 hours of volume. All are accessible by pumper truck if required.

5.1.4 Time lines for detection of each trigger event type, in hours, taking into account best and worst case scenarios (during and after working hours including weekends)

1. Detection of a trigger event (Section 5.1) is 0.25 hours in normal working hours and up to 4 hours after normal working hours.

5.2 Notification

Any trigger event described in Section 5.1 requires a notification of the event by the Ladysmith WWTP Chief Operator or designate as follows:

2. The Ladysmith WWTP Chief Operator or designate immediately notifies both the EMBC and the DFO Radio Room verbally at the phone numbers noted in Appendix D when trigger events are detected as per Section 5.1, noting the event type, the estimated start time and name of the caller as well as when the event has ceased. As of April 1, 2022 in the event that a trigger event occurs outside of regular business hours (Monday-Friday between the hours of 08:00-16:30 and statutory holidays) the DFO standby officer on duty is to be immediately notified, in replacement of the DFO Radio Room. A standby calendar containing the name and contact information for the officer on standby has been provided in the event of any after-hours emergencies (Appendix D). (0.50 hours)

Note: A notification under the current CMP does not replace or otherwise affect the requirements, pursuant to subsection 38(4) of the *Fisheries Act (R.S.C., 1985, c. F-14)*, to report to a *Fisheries Act (R.S.C., 1985, c. F-14),* inspector or to any other person or authority as is prescribed by regulations, when there occurs a deposit of a deleterious substance out of the normal course of events in water frequented by fish, or a serious and imminent danger thereof. A notification does not replace the emergency procedures and bypass requirements as defined under operational certificate 110161.

3. The Ladysmith WWTP Chief Operator or designate enters all information into the WWTP Reporting Tool, when a trigger event is detected or has ceased. (0.50 hours)

   Also notify of any planned or unplanned changes in operations at the Ladysmith WWTP (e.g. untreated sewage discharge, proposed maintenance work, etc.) which are likely to alter the normal effluent loading or location of discharge in or in proximity of the conditionally classified areas herein described.

   Upon submitting the event record into the WWTP Reporting Tool, a "WWTP Event Report" (Appendix C) will automatically be distributed via email (subject title "NOTICE: WWTP Start of Event") to the Participants listed in this CMP (Appendix D).

4. When the trigger event has been terminated and the operation of the Ladysmith WWTP has returned to normal, the WWTP Chief Operator or designate will update the event record in the WWTP Reporting Tool, which will automatically distribute an updated "WWTP Event Report" via email (subject title "NOTICE: WWTP End of Event") to each Participant listed in this CMP (Appendix D).

   If the WWTP Reporting Tool is not working as expected, the WWTP Chief Operator or designate should report the problem to the DFO CSSP Coordinator (Appendix D) as soon as possible, providing the source location and start time of the discharge.

   The WWTP Chief Operator or designate will follow the instructions described in section "7.3 The WWTP Reporting Tool is Offline or Malfunctioning" for an alternate means of notifying the Participants of a trigger event. These instructions are found in section "7. Trouble Shooting" in the "User Manual – Wastewater Treatment Plant Reporting Tool" document (Appendix C).

5.2.1 Description of how notification is provided to all CSSP partners and other Participants (phone/fax/email)

5. The EMBC completes and sends a Dangerous Goods Incident Report (DGIR) immediately and notifies other agencies including ECCC, ENV and DFO/CCG Marine Communications and Traffic Services (MCTS) by sending the incident report by email (Appendix C). (0.25 hours)

6. The DFO/CCG MCTS notifies the DFO Radio Room, or standby officer after business hours, by also sending the DGIR report (built in redundancy). (0.25 hours)

7. The DFO Radio Room notifies the CSSP Coordinator and/or DFO Conservation and Protection during business hours. The DFO standby officer will be contacted directly after business hours. (0.25 hours)

5.3 Response

Upon receiving notification as outlined in Section 5.2, each Participant will respond in accordance with their respective authorities as follows:

8. The DFO CSSP Coordinator and/or C&P, contact the CFIA Pacific Shellfish Desk by email at pacificshellfish@inspection.gc.ca with the subject line: **URGENT - WWTP OVERFLOW EMERGENCY NOTIFICATION.** Outside of business hours the DFO standby officer will contact CFIA directly by phoning the Pacific Shellfish Desk and/or Timothy Delange at the numbers listed in this CMP (Appendix D). (0.25 hours)[1]
9. Once the DFO CSSP Coordinator or standby officer reaches the Pacific Shellfish Desk or Timothy Delange outside of business hours, the CFIA will email all registered molluscan shellfish processing plants immediately. (0.25 hours)[1]
10. DFO (CSSP Coordinator or C&P) are responsible for (1.0 hours);
    a. DFO will initiate their internal procedures regarding the activation of a change in area status through SHELLI (Shellfish Harvest Extent, Latitude, Longitude Information) which will invoke a Prohibition Order to place the area in Closed status under the *Management of Contaminated Fisheries Regulations (SOR/90-351)* (notification that provides DFO Regional Director General with reason to believe that fish of any species are contaminated), C&P Regs will accept the recommendation and a Prohibition Order will be sent to the RDG to be signed.
    b. Inform affected harvesters (First Nations, commercial harvesters and stakeholders) (Appendix D) via email,
    c. Inform CSSP partners of overflow or termination of an overflow,
    d. Post Fishery Notice via Automatic Fishery Notice System,
    e. Update public communication material (http://www.pac.dfo-mpo.gc.ca/fm-gp/contamination/index-eng.html) with a map and the legal description of the area affected through FRIS,
    f. Initiate patrols of affected area.

[1] Steps 8-9 occur simultaneously. The maximum time allotted to each step is 0.25 hour and are not additive.

The total detection, notification and response time is **7.0 hours[1]**.

[1] **Note:** The total detection, notification and response time is calculated by adding together the total time allotted for steps 1 to 10 (7.0 hours).

11. Upon submission of an event report into the WWTP Reporting Tool, a notification will automatically be distributed via email on behalf of DFO (subject title “NOTICE: WWTP

Discharge") to the Participants stating the area has been affected by a trigger event and the issuance of a prohibition order and notice of the order will be published no later than the next business day to inform harvesters to cease harvesting immediately in the area subject to the order. This notification will be distributed within one hour of detection of the discharge.

12. Upon submission of an event report into the WWTP Reporting Tool, a notification will automatically be distributed via email on behalf of CFIA (subject title "NOTICE: WWTP Discharge") stating the area has been affected by a trigger event and that the conditionally classified area(s) have been closed or are about to be closed to harvesting. This notification will be distributed within one hour of detection of the discharge.

6 RE-OPENING CRITERIA

The harvest areas described in this CMP will remain in closed status to harvesting until the re-opening criteria are met. Commencement of re-opening criteria does not begin until after the Town of Ladysmith has provided notification that the trigger event(s) have ceased as outlined in Section 3 and Section 5.

Areas will be returned to their classification status when conditions outlined in Section B, 4 of the CSSP Manual have been met as advised by ECCC and CFIA.

The sampling methodology must comply with the established procedures outlined in Section B, 7 of the CSSP Manual and the Sampling Policy and Procedures found in the CFIA's Fish Products Standards and Methods Manual. The sample locations are identified on a map in Appendix A.

All laboratories performing CSSP testing for regulatory purposes must be accredited to the international standard ISO/IEC 17025 "General Requirements for the Competence of Testing and Calibration Laboratories" by a recognized Canadian accrediting body. All samples taken in support of the CMP area are submitted to an ISO/IEC 17025 accredited laboratory.

6.1 In addition to the conditions above, the following arrangement has been reached between the signatories to this CMP as to the process and responsibilities for collecting samples, sample locations, and where they are analyzed.

- The area must remain in closed status for a minimum of 7 days after the latest trigger event has ceased.
- In order to re-open sooner than 21 days after the latest trigger event or once a sewage bypass over 50 m^3/day has ceased, ECCC and CFIA trained samplers may collect water and shellstock samples from established verification stations outlined in Appendix A. Water and shellstock should be collected no earlier than 5 days after the most recent trigger event has ceased.
- The sampling methodology must comply with the established procedures outlined in Section B, 7 of the CSSP Manual and Sampling Policy Procedures found on the CFIA's website at http://inspection.gc.ca/food/requirements-and-guidance/preventive-controls-food-businesses/sampling-procedures/eng/1518033335104/1528203403149. The sample locations are identified on a map in Appendix A.

- Only those samplers operating under a current and valid sampler agreement with ECCC may perform water verification sampling.

ECCC and CFIA will make a recommendation to DFO to revoke the closure and the harvesting prohibition:

- When a minimum of 21 days have elapsed following the termination of the most recent trigger event, or
- When marine water and shellstock samples from the conditional areas are confirmed to meet the standards for harvest set forth in the CSSP, and a minimum of 7 days have elapsed following the termination of the most recent trigger event.

7 ANNUAL REPORTING

The Participants will provide input into an annual report on the management of the area. The report will then be provided to the Pacific Region Interdepartmental Shellfish Committee (PRISC) each spring for review. This report shall include, as a minimum, the information outlined in Appendix B.

7.1 **Procedures to be followed at the local level in order to complete the report:**

DFO will lead the development of an Annual Report for Ladysmith.

Summary of activities will include detailed information about each failure detection, notification and response, including timelines, action and delays during the chain of events leading to closure and notification to First Nations, stakeholders and the public and subsequent openings together with all supporting documentation. DFO will provide details on the timelines from detection to closure; as well as a summary of surveillance, enforcement, and control activities: number of patrols, number of incidents, violations.

The CMP annual report shall be completed for review and accepted by a PRISC working group by May 1st of the following year.

CFIA and ECCC will provide input to the Annual Report with water and shellstock microbiological data used to re-open the area (dates, results). Summary data is required by Feb 1st of the following year.

ENV will provide an annual report of the results of activities listed under the ENV in Section 3 in relationship to the CMP by Feb 15th of the following year including but not limited to any recommendations for changes to the CMP.

A report from the Town of Ladysmith /WWTP Chief Operator or designate will include a summary of the occurrence trigger events including, dates, estimates of discharges and records of the notifications made during the duration of this management plan. The report shall be submitted to ENV by January 15th of the following year.

Concerns and recommendations may be provided by all Participants and included in the Ladysmith CMP Annual Report.

8 AMENDMENT AND TERMINATION

Any Participant may, upon providing written notice to the other Participants, withdraw from this voluntary CMP.

If at any time any Participant to the CMP fails to fulfill the requirements as set forth in the CMP or gives notice of withdrawal, the Pacific Region Interdepartmental Shellfish Committee (PRISC) will determine whether the area classification or status will be changed.

This CMP may be amended at any time subject to the written approval of all the Participants.

9 APPENDICES

The Appendices herein form part of this CMP.

Appendix A – Maps:

Figure 1. Ladysmith WWTP Area Classification and Classification Boundaries;

Figure 2. Water and shellstock sample collection locations

Appendix B – Sample Annual Report – Information for the Report

Appendix C – Wastewater Treatment Plant Event Report and Discharge of Wastewater Notice - WWTP Event Report

Appendix D – Contact List

Appendix E – CSSP and Conditional Management Plan Definitions

Approved at ____________________, this _______ day of ______________, 2022.

____
Rebecca Reid
Regional Director General
Fisheries and Oceans Canada
Pacific Region

____
Kelvin Mathuik
Director General
Western Operations
Canadian Food Inspection Agency

____
Arash Shahsavarani
Executive Director, Water Quality Monitoring and Surveillance Division
Environment and Climate Change Canada

____
Laurel Nash
Assistant Deputy Minister, the Province of British Columbia Ministry of Environment and Climate Change Strategy, Environmental Protection Division (ENV)

____
Director of Corporate Services (or designate)
Town of Ladysmith

Appendix A –*Maps*

Shellfish harvesting will become prohibited in conditionally restricted area (Figure 1) when an effluent discharge meeting the criteria of a trigger event is reported. Further guidance of how licensed shellfish processing establishments maintain control of shellfish harvested from areas situated between the prohibited area and the response line is contained in CFIA policy documents found at http://www.inspection.gc.ca/food/sfcr/general-food-requirements-and-guidance/preventive-controls-food-businesses/fish/live-shellfish/eng/1515437226516/1515437308440.

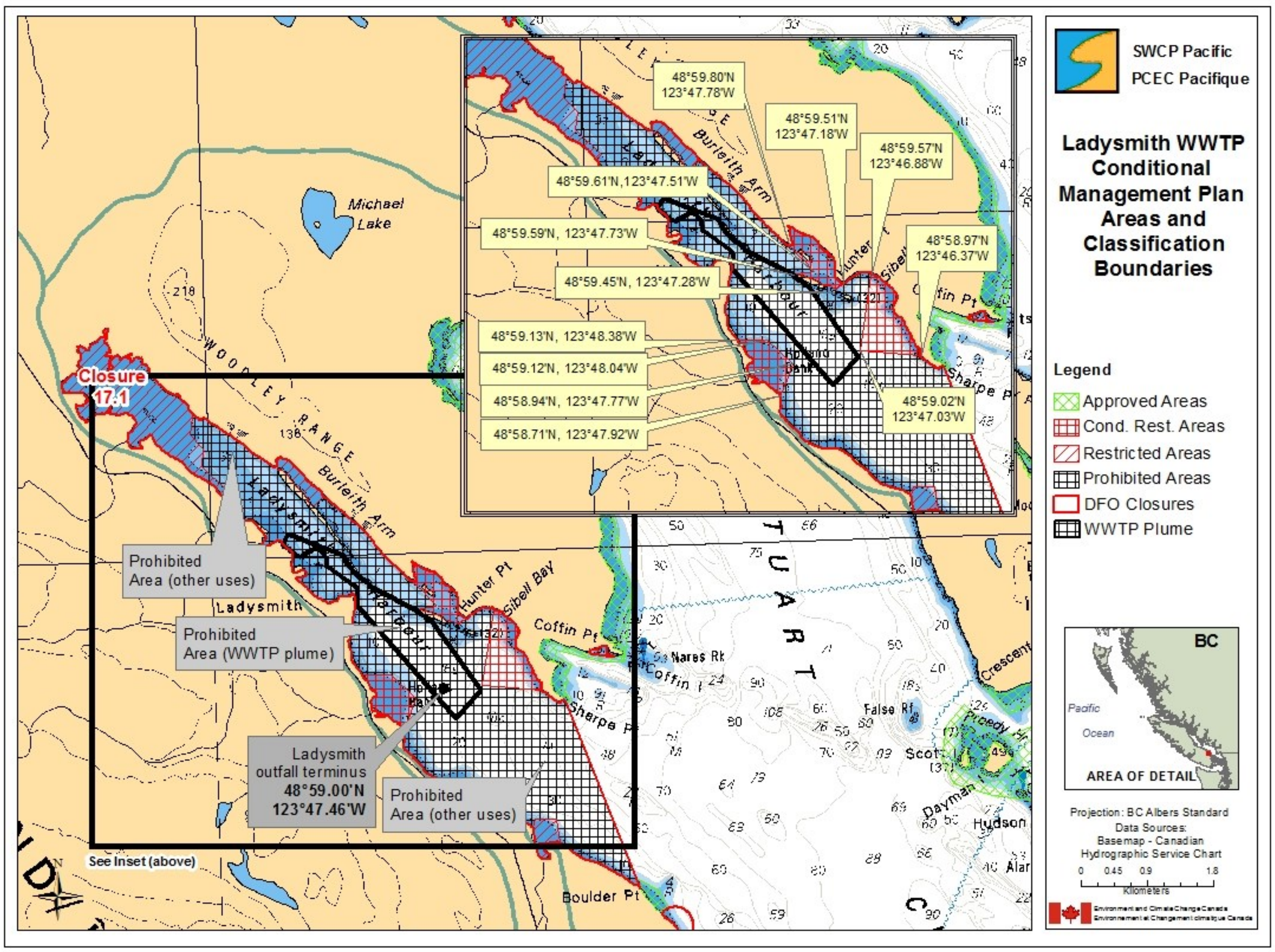

Figure 1. Ladysmith WWTP Area Classification and Classification Boundaries.

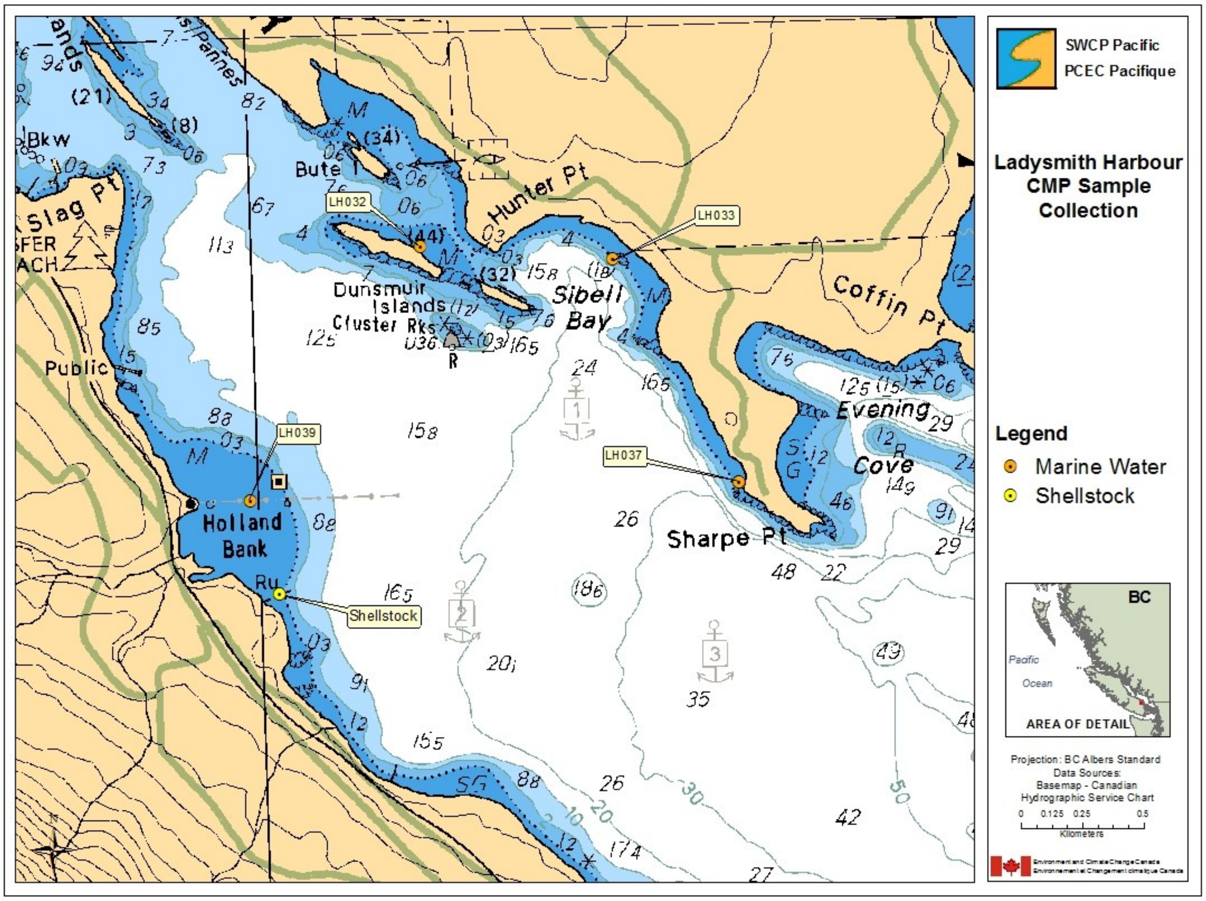

Figure 2. Water and shellstock sample collection locations.

Appendix B – Sample Annual Report – Information for the Report

Name of Area
Conditional Shellfish Area Annual Report for (insert year)

Area

Description/location with boundaries

Map (with classification and sampling sites for water quality and shellstock)

Closure criteria

Potential time period for opening (if applicable)

Species managed and harvesting restrictions/season (if applicable)

Summary of Activities

Number of openings/closures during the year

Prohibition order numbers and dates

Supporting documentation used to make decision about closing
Notices from WWTP Chief Operator or designate (Event, dates, duration)

Supporting documentation used to make decision about opening
Water and shellstock microbiological data to re-open the area (dates, results)

Surveillance, enforcement, control activities: number of patrols, number of incidents, violations

Reference to Management Plan

http://www.pac.dfo-mpo.gc.ca/fm-gp/ifmp-eng.html

Report from Province

Comments on the WWTP annual report, as a compliance review to confirm that trigger events were detected and responded to as described in the CMP.

Report from Municipality/WWTP Chief Operator or designate

Summary report of discharges and notifications

Concerns/ Recommendations (all Participants)

Appendix C – WWTP Reporting Tool Event Report Procedures

https://docs.google.com/document/d/1G6Sq6cg26L1_jkeLQJSMhWh5g28KSxt1/edit?usp=sharing&ouid=115899308216834663229&rtpof=true&sd=true

Example of a "Discharge of Wastewater Notice WWTP Event Report"

SECTION 1: LOCATION & REPORTING

Facility Name:		Initial Reporting Operator:	
Address:		DGIR No.	
		Date of Initial Report:	
Permit No.		Time of Initial Report:	

Initial Contact/Reporting:

- ☐ EMBC (Emergency Management BC)
- ☐ DFO Radio Room
- ☐ Limberis Seafood Processing

Follow-up Report Submission

- ☐ ECCC
- ☐ CFIA
- ☐ DFO
- ☐ ENV
- ☐ Ladysmith Manager of Operations

SECTION 2: DESCRIPTION OF EVENT

☐ WWTP disruption that results in more than 50m3/d of untreated effluent being discharged via the bypass outfall	☐ non-planned Emergency ☐ Planned Maintenance
☐ WWTP disruption that results in untreated or partially treated effluent discharge that fails to meet provincial authorized disinfection requirements	☐ non-planned Emergency ☐ Planned Maintenance
☐ Daily effluent discharge exceeding 14,400m3/day	☐ non-planned Emergency ☐ Planned Maintenance
☐ Collection system failures that result in an untreated effluent discharge to the marine environment	☐ non-planned Emergency ☐ Planned Maintenance
☐ Spirogester or MBBR overflow (primary treated, in receipt of disinfection) greater than 50m3/day	☐ non-planned Emergency ☐ Planned Maintenance

Specific Description/Comments:

SECTION 3: BY-PASS DETAILS

Date By-passing Started:		Date By-passing Stopped:	
Time By-passing Started (50m³):		Time By-passing Stopped:	
Volume (m³):	☐ Estimated ☐ Measured	Duration of By-pass:	

SECTION 4: ACTIONS TAKEN TO CONTROL OR REMEDIATE EVENT

Report Prepared By/Position:	
Supervisor Review By/Position:	
Submitted By:	

Date:	
Time:	

Appendix D –Contact List

DEPARTMENT/ AGENCY	NAME	POSITION	TELEPHONE/ FAX	EMAIL
Fisheries and Oceans Canada	Erin Milligan	Resource Management CSSP Coordinator	Cell: (250) 327-4606	erin.milligan@dfo-mpo.gc.ca
	Greg Plummer	Conservation and Protection	Tel: (250) 286-5815 Cell: (250) 830-4478	greg.plummer@dfo-mpo.gc.ca
	Carla Neumann (1st alternate)	Conservation and Protection	Cell: (250) 286-5820	carla.neumann@dfo-mpo.gc.ca
Canadian Food Inspection Agency	Timothy Delange	Supervisor, Inspection and Advisory Services	Cell: (250) 951-8154	timothy.delange@inspection.gc.ca
	Gerry Morello (1st alternate)	Senior Inspector	Tel: (250)213-2519 Fax: (250) 363-0144	gerry.morello@inspection.gc.ca
	Pacific Shellfish Desk	Inspection Advisors	Cell: (236) 330-2967 Cell: (604) 418-4363	pacificshellfish@inspection.gc.ca
Environment and Climate Change Canada	Elizabeth Graca	Regional Head, Shellfish Water Classification Program- Pacific	Tel: (604) 903-4475 Cell: (604) 209-5810	elizabeth.graca@ec.gc.ca
	Paul Moccia	Area Coordinator	Tel: (604) 903-4425 Fax: (604) 903-4423	paul.moccia@ec.gc.ca
BC Ministry of Environment and Climate Change Strategy	Stephanie Little	Section Head Compliance	Tel: (250) 490-8258 Cell: (778) 622-6908	stephanie.little@gov.bc.ca
	Cassandra Caunce	Regional Director, South Authorizations	Tel: (604) 930-7119 Cell: (236) 468-2227	cassandra.caunce@gov.bc.ca
Town of Ladysmith, Director of Infrastructure Services	Ryan Bouma	Director of Infrastructure Services	Tel: (250) 245-6440 Cell: (250) 714-9235	rbouma@ladysmith.ca
Town of Ladysmith WWTP Chief Operator	Mike Brown	Supervisor, Treatment and Supply	Tel: (250) 924-1302 Cell: (250) 713-3165	mbrown@ladysmith.ca
Stz'uminus First Nation	Roxanne Harris	Chief and Acting Director of Administration	Tel: (250) 245-7155 Fax: (250) 245-3012	chief@stzuminus.com
Stz'uminus First Nation	Teoni Jameson	Acting Senior Community Manager	Tel: (250) 245-7155 Ext: 282 Fax: (250) 245-3012	teoni.jameson@stzuminus.com
Stz'uminus First Nation	Krista Perrault	Emergency Coordinator	Cell: (250) 739-1468 or (250) 618-4257	krista.perrault@stzuminus.com
Limberis Seafood Processing Ltd.	Leo Limberis	Depuration/relay licence holder	Tel: (250) 245-3021 Fax: (250) 245-3606	kath@limberisseafood.com
Emergency Management BC			1-800-663-3456	
DFO Radio Room			1-800-465-4336	

Appendix E – CSSP and Conditional Management Plan Definitions

Approved Area - The classification assigned to a shellfish harvest area as determined by the shellfish control authority from which shellfish can be harvested for direct consumption.

Bypass – Effluent flow at a wastewater treatment plant or lift station that bypasses the treatment plant and is discharged to the marine environment. Also the system that directs the effluent flow to bypass the treatment plant.

Canadian Shellfish Sanitation Program - A program to classify and monitor shellfish harvest areas to determine whether shellfish are safe for human consumption and to regulate harvesting from those areas.

Conditional Management Plan (CMP) - An agreement signed by relevant parties for the management of shellfish harvest in conditionally classified areas.

Conditionally Restricted Area – The classification assigned to a shellfish harvest area which has been determined by the shellfish control authority to meet, at a minimum, the restricted classification criteria for a predictable period. The period is conditional upon meeting established requirements and/or performance standards specified in a conditional management plan.

Controlled Purification or Depuration - The process of using a controlled, aquatic environment in a depuration establishment to reduce the level of microbiological contamination in live shellfish.

Detection: The point in time at which a wastewater treatment or collection system release / discharge incident (as defined in the relevant CMP) is first detected by a wastewater treatment plant operator or collection system operator (or delegate).

Emergency Closure - A shellfish harvesting area in the open status may be placed in the closed status via an emergency closure when it is suspected that shellfish may become contaminated as a result of a temporary emergency situation. Emergency situations may include natural or operational events such as severe storms, flooding, and spills of oil, toxic chemicals or significant sewage discharges.

Lift Station – Part of a wastewater treatment plant collection system.

Moving Bed Biofilm Reactor (MBBR) – A type of wastewater treatment process that consists of an aeration tank with special devices that provide a surface where a biofilm can grow. The aeration system of the tank must continually mix the influent and the devices covered in the biofilm in order to provide treatment to the wastewater.

Notification – The point in time at which one of the federal CSSP partners receives notice from a wastewater treatment or collection system operator (or delegate) of a trigger event (as defined in the relevant CMP).

Prohibited Area - The classification assigned to a shellfish harvest area as determined by the shellfish control authority where shellfish harvesting is not permitted.

Prohibition Order – A regulatory mechanism used by Fisheries and Oceans Canada to close and open shellfish harvesting areas for fishing bivalve shellfish.

Response - A series of actions taken by the federal shellfish control authorities as defined in the relevant CMP based on the classification of the area that will serve to ensure that product does not reach market and the implicated area is placed in closed status.

Response Line - The boundary at which the sewage effluent plume is predicted to lie during a wastewater treatment plant or collection system trigger event when the competent shellfish control authority will respond.

Restricted Area - The classification assigned to a shellfish harvest area as determined by the shellfish control authority where harvesting shall be by license under the *Management of Contaminated Fisheries Regulations* and the shellfish, following harvest, is subjected to a suitable and effective treatment process through relaying or depuration.

Shellfish Control Authority – The departments or agencies of the Government of Canada that are signatories to the interdepartmental Memorandum of understanding between the CFIA and DFO and ECCC concerning the CSSP or provincial shellfish leasing bodies.

Shellstock – Live shellfish in the shell.

Status - Describes whether shellfish harvest is permitted and is independent of the classification of the area[1]

- **Open** - Any classified area where shellfish harvest is authorized.
- **Closed** - Any classified area where shellfish harvest is not authorized.
  There may be circumstances under which areas in closed status can be harvested for depuration or relay under MCFR (*Management of Contaminated Fisheries Regulations)* license provided that the requirements for such a license are met.

Trigger Event – Any event or disruption that results in untreated effluent or effluent with insufficient or inadequate disinfection being released from the WWTP that causes action to commence closure of an area to shellfish harvest.